Calibration-Gated Combinatorial Bandits with Untrusted LLM Pool-Builders

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Abstract

Large language models are increasingly deployed as advisors that propose shortlists of candidate actions for downstream decision algorithms. We study this setting formally as a combinatorial semi-bandit problem in which an unreliable LLM acts as a *pool-builder*, returning a candidate subset of βm arms from d that the bandit explores within. Existing LLM-augmented bandit algorithms either trust the oracle through self-consistency signals or treat its output as a fixed prior. We show that any algorithm whose oracle-influence is gated only by self-consistency suffers $\Omega(T)$ regret against deterministic adversaries, a failure mode that we call "consistent-wrong" corruption. The natural trust-based combinatorial method incurs $2.3\times$ to $4.2\times$ worse regret than vanilla CUCB under such oracles, and naive trust methods such as LLM-Greedy and OPRO-Bandit incur up to $4.2\times$ worse regret. A pool-distillation baseline alone (POOL-CTS) fails even more severely on large-gap instances, accumulating regret over $60\times$ that of vanilla CTS. We introduce Pool-CTS-CG, an algorithm that distills the LLM into a candidate pool, runs combinatorial Thompson sampling restricted to the pool, and uses a soft sigmoid calibration gate that reads the empirical reward signal to mix in full-action-space exploration when pool quality is in doubt. We prove that Pool-CTS-CG achieves the pool-restricted regret rate when the LLM is well calibrated and recovers the unrestricted Thompson sampling rate when it is adversarial. Across five corruption regimes and three gap structures we observe that Pool-CTS-CG matches or beats the strongest non-LLM baseline in every reliable-oracle scenario while reducing the consistent-wrong worst-case regret of pool-only methods by at least an order of magnitude.

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1. Introduction

Large language models are now common as decision advisors in production systems. A typical pattern in industry is the LLM-pool-builder: a single LLM call returns a shortlist of candidate items for a recommender, an ad slate, a tool selection layer, or an experiment design (Liu et al., 2024; Bayley et al., 2025; Alamdari et al., 2024). A downstream learner then explores within the proposed pool to identify the highest-reward subset under feedback. The LLM never observes rewards directly. The bandit algorithm never sees more than the proposed pool.

This deployed pattern is missing from the bandit literature. Existing LLM-bandit work studies either the LLM as the bandit itself (Krishnamurthy et al., 2024; Nie et al., 2024), the LLM as an offline prior (Bayley et al., 2025; Alamdari et al., 2024), or the LLM as a contextual feature extractor. None of these match the pool-builder pattern, and none stress-test the failure modes that are most likely to occur in deployment. Specifically, prior empirical studies inject independent and identically distributed noise into LLM outputs to simulate "unreliable" oracles, while the failure modes that arise from prompt drift, distribution shift, jailbreaks, or systematic bias produce *consistently wrong but plausible* suggestions. We refer to this as the consistent-wrong regime.

We make the following contributions.

(C1) Lower bound on consistency-only trust. We prove that any algorithm whose use of the LLM oracle is gated solely through a self-consistency statistic suffers regret $\Omega(T\Delta_{\min}m/d)$ against deterministic adversaries (Theorem 1). The self-consistency statistic that drives the most prominent LLM-augmented bandit method is therefore provably circumventable.

(C2) An algorithm that does not fall into the consistency trap. We propose Calibration-Gated Pool Thompson Sampling (Pool-CTS-CG), Algorithm 1. After a brief round-robin exploration phase to produce baseline empirical reward estimates, the algorithm constructs a pool from K

LLM queries and runs combinatorial Thompson sampling. A soft sigmoid gate compares the pool-restricted top-m empirical reward against the unrestricted top-m empirical reward and routes each round between pool sampling and full sampling probabilistically. The mechanism reads the bandit's reward signal rather than oracle-only statistics, so it is not gated solely by self-consistency and the lower bound of (C1) does not apply.

(C3) Two-regime regret bound. Theorem 2 shows that Pool-CTS-CG achieves regret $\tilde{O}((\beta m)^2 \log T/\Delta_{\min})$ when the pool covers the optimal super-arm, and recovers the standard combinatorial Thompson sampling rate $\tilde{O}(m^2 \log d \log T/\Delta_{\min})$ when the pool excludes it. The transition between regimes is smooth and is governed by a single hyperparameter, the gate scale σ_g.

(C4) Empirical validation across the corruption taxonomy. We benchmark Pool-CTS-CG against ten baselines, including the dominant LLM-augmented bandit methods, across five corruption regimes (perfect, uniform, consistent-wrong, partial overlap, adversarial) and three gap structures (uniform, clustered, graded). Our method dominates all LLM-augmented baselines on average regret across the corruption taxonomy and is the only method that simultaneously beats the non-LLM baseline on reliable oracles and matches it on adversarial oracles. The consistent-wrong worst-case regret of pool-only baselines is reduced by up to $60\times$ in the clustered environment.

2. Related Work

Combinatorial semi-bandits. The combinatorial multi-armed bandit framework was introduced by Chen et al. (2013) (CUCB), who established gap-dependent regret rates for general non-linear reward functions through approximation oracles. Combinatorial Thompson Sampling (Wang & Chen, 2018) achieves the matching $\tilde{O}(m \log K_{\max} \log T/\Delta_{\min})$ rate using Beta posteriors. Recent work extends these results to graph feedback (Wen et al., 2025) and offline pretraining (Liu et al., 2025).

Corruption-robust bandits. Lykouris et al. (2018) introduced the adversarial-corruption model with budget C for stochastic bandits, achieving regret $O(K \log T/\Delta_{\min} + KC)$. Refinements include Gupta et al. (2019) for tighter dependence and He et al. (2022) for linear contextual bandits. Fang et al. (2025) introduce BARBAT, an epoch-based framework that achieves $\tilde{O}(\sqrt{KT} + C)$ matching the lower bound. None of these works address *action-space* corruption, where the corruption affects which actions are proposed rather than the rewards observed.

LLM-augmented bandits. Krishnamurthy et al. (2024) stress-tested in-context exploration by GPT-4 and found that without explicit scaffolding the LLM exhibits suffix failures and rarely achieves sublinear regret. Nie et al. (2024) fine-tune small models on UCB and TS trajectories to recover algorithmic exploration behavior. Liu et al. (2024) use LLMs as warm-starters and surrogates inside Bayesian optimization. Alamdari et al. (2024) use LLM-generated pseudo-data to initialize a bandit posterior but do not stress-test the prior. Bayley et al. (2025) document that LLM-initialized bandits break under noisy priors above approximately 30% corruption but do not provide a corrective algorithm. The closest work in spirit is Cao et al. (2026), which uses an LLM advisor with a robustness guarantee, but in the contextual rather than combinatorial setting.

Learning-augmented algorithms. The robustness-consistency frontier framed by Lykouris & Vassilvitskii (2018) is conceptually similar: an algorithm should match a strong rate when the prediction is good and degrade gracefully when it is bad. Our calibration gate operationalizes this in the combinatorial bandit setting and the pool-builder LLM role.

Concurrent work. A concurrent line of work, exemplified by Anonymous (2025), applies calibration-style gating to LLM pseudo-observations in the *single-arm contextual* bandit setting and shows empirical gains on news recommendation. Our work differs in three respects. First, we study the *combinatorial semi-bandit* setting in which the LLM proposes a super-arm pool of size βm, not a single context-conditioned reward estimate. Second, we provide a corruption-robust regret guarantee (Theorem 2) and a matching consistency-trap lower bound (Theorem 1); their work is empirical. Third, their gate operates as an exponential-moving-average decay on pseudo-observation weights; our gate is a soft mixture between pool-restricted and full-arm Thompson sampling, a different mechanism that targets the dimension-reduction benefit specific to combinatorial actions.

3. Setup

Problem. A learner faces a ground set of d arms over T rounds. A super-arm $S \subseteq [d]$ with $|S| = m$ is played each round. Each base arm $i \in [d]$ has unknown mean $\mu_i \in [0, 1]$. The reward of S at round t is $r_t(S) = \sum_{i \in S} X_{t,i}$, where $X_{t,i} \sim \text{Bernoulli}(\mu_i)$ is observed for each $i \in S$ (semi-bandit feedback). Let $S^\star = \arg\max_{|S|=m} \sum_{i \in S} \mu_i$ and $\Delta_{\min} = \min_{i \in S^\star, j \notin S^\star} (\mu_i - \mu_j)$. Cumulative pseudo-regret is $R(T) = T \sum_{i \in S^\star} \mu_i - \mathbb{E}[\sum_{t=1}^{T} r_t(S_t)]$.

LLM oracle. The learner has access to a stochastic oracle $\mathcal{O}$ that, on query, returns a candidate super-arm $\tilde{S} \subseteq [d]$ with $|\tilde{S}| = m$. The oracle distribution is unknown to the learner. The total number of oracle queries across the run

is bounded by a constant K that does not scale with T. We do not assume the oracle is calibrated, unbiased, or even consistent across queries.

Corruption taxonomy. Following the taxonomy that emerges from the deployed-LLM literature, we identify four corruption regimes that the oracle may exhibit, each parameterized by a corruption rate $\epsilon \in [0, 1]$.

- **Uniform**: with probability ϵ the oracle returns a uniformly random super-arm; otherwise it returns $S^\star$.
- **Adversarial**: with probability ϵ the oracle returns the worst-m arms (smallest means); otherwise $S^\star$.
- **Consistent-wrong**: the oracle deterministically returns the best-m *suboptimal* arms (high mean but not in $S^\star$).
- **Partial overlap**: the oracle returns a set with $(1-\epsilon)m$ correct and ϵm incorrect arms.

The consistent-wrong regime is the empirically most damaging one and is the focus of Theorem 1. It models the realistic failure case in which an LLM has been miscalibrated through prompt drift or adversarial fine-tuning to produce plausible but suboptimal suggestions.

Pool-builder protocol. The learner queries the oracle a small number of times during an initial pool construction phase. Let $\mathcal{P} \subseteq [d]$ denote the resulting candidate pool with $|\mathcal{P}| = \beta m$ for a constant $\beta > 1$. After pool construction, the learner does not query the oracle further. All rounds use the bandit's own selection rule, optionally restricted to $\mathcal{P}$.

4. Algorithm

We propose Pool-CTS-CG, summarized in Algorithm 1. The algorithm has three phases. Phase 1 (lines 1 to 4) performs a round-robin exploration over the full ground set for T_{init} rounds. This produces an empirical reward estimate $\hat{\mu}$ for every arm, including arms that will not be admitted to the pool. Phase 2 (lines 5 to 7) queries the oracle K times, aggregates the suggestions by frequency, and constructs a pool $\mathcal{P}$ of size βm together with n_s random safety arms. Phase 3 (lines 8 to 17) runs combinatorial Thompson sampling. At each round, the algorithm computes a calibration gate $g_t \in [0, 1]$ that compares the empirical top-m on the full ground set against the empirical top-m restricted to the pool. The round routes to full-action-space sampling with probability g_t and to pool-restricted sampling with probability $1 - g_t$.

The gate g_t is a sigmoid function with scale σ_g, where smaller σ_g produces sharper transitions. The argument to the sigmoid is the difference between the unrestricted top-m reward sum and the pool-restricted top-m reward sum, both computed under the current empirical mean $\hat{\mu}$. When the pool contains the optimal super-arm, this difference is small or zero and $g_t \approx 0$, so the algorithm runs pool-restricted CTS and benefits from the reduced effective dimension. When the pool excludes the optimal arms, the unrestricted top-m identifies them as the empirical mean accumulates, the difference grows, and $g_t \to 1$, routing to full Thompson sampling.

Algorithm 1 Pool-CTS-CG: Calibration-Gated Pool-CTS

Require: Ground set $[d]$, super-arm size m, pool size βm, safety arms n_s, oracle $\mathcal{O}$, queries K, init horizon T_{init}, gate scale σ, recompute period τ.

1: *Phase 1: Round-robin exploration.*
2: **for** $t = 0, 1, \ldots, T_{\text{init}} - 1$ **do**
3: Play super-arm $S_t = \{(t \cdot m + j) \bmod d : j = 0, \ldots, m-1\}$ and observe rewards.
4: **end for**
5: *Phase 2: Pool construction.*
6: Query oracle K times to obtain $\tilde{S}^{(1)}, \ldots, \tilde{S}^{(K)}$. Let f_i denote the frequency of arm i across queries.
7: Let $\mathcal{P}$ be the βm arms with largest f_i, augmented with n_s uniformly random arms from $[d]$.
8: *Phase 3: Calibration-gated CTS.*
9: Initialize Beta posteriors $(\alpha_i, \beta_i) = (1, 1)$ for $i \in [d]$.
10: **for** $t = T_{\text{init}}, T_{\text{init}} + 1, \ldots, T - 1$ **do**
11: **if** $(t - T_{\text{init}}) \bmod \tau = 0$ **then**
12: $g \leftarrow \sigma\left(\frac{\sum_{i \in \text{Top}_m(\hat{\mu})} \hat{\mu}_i - \sum_{i \in \text{Top}_m(\hat{\mu} \mid \mathcal{P})} \hat{\mu}_i}{\sigma_g}\right)$
13: **end if**
14: Sample $\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$ for $i \in [d]$.
15: With probability g, set $S_t \leftarrow \text{Top}_m(\theta)$.
16: Otherwise set $S_t \leftarrow \text{Top}_m(\theta \mid \mathcal{P})$.
17: Play S_t, observe $\{X_{t,i}\}_{i \in S_t}$, update (α_i, β_i).
18: **end for**

Why a soft gate. A natural alternative is a hard threshold: keep the pool if and only if the agreement statistic exceeds some cutoff. We tested this design (referred to as Pool-CTS-IC in our experiments) and observed bimodal performance under consistent-wrong corruption: detection succeeded on roughly half the random seeds and failed on the rest, producing standard deviations of several thousand in regret. The bimodality is intrinsic to threshold-based detection at finite samples. The sigmoid gate replaces the discrete decision with a smooth mixture, which both eliminates the bimodality and provides a graceful degradation across calibration regimes.

The gate is a generic mechanism. The sigmoid is one specific instance of a broader class of soft gates. Any monotone-decreasing, L-Lipschitz function $g : \mathbb{R} \to [0, 1]$ of the cal-

ibration statistic $\Delta_t = \sum_i \hat{\mu}_{(i)} - \sum_i \hat{\mu}_{(i)|\mathcal{P}}$ produces the same two-regime regret guarantee of Theorem 2 up to a constant multiplicative factor that depends on L. We adopt the sigmoid because it has a single interpretable scale parameter and admits a closed-form derivative; the conclusions of Section 6 carry over to other reasonable choices.

Why pool size $\beta = 3$. The pool-size multiplier β is set so that $|\mathcal{P}| = \beta m$ comfortably exceeds the super-arm size m but stays small relative to d. The default $\beta = 3$ matches the size used in standard top-k recommendation pipelines (slates of 5 items with $m \in \{3, 5\}$) and in the related contextual-bandit pool work of Anonymous (2025). The bound of Theorem 2 holds for any fixed $\beta \in [1, d/m]$ with the leading term scaling as $(\beta m)^2$; we therefore prefer the smallest β that empirically maintains pool coverage above 95% across the corruption regimes we test, which is $\beta = 3$ for our default $d = 100$, $m = 10$.

5. Theory

Notation. Let $\mathcal{P}$ denote the pool returned by Phase 2 of Algorithm 1, and let the *pool-coverage* indicator be $Z = \mathbf{1}\{S^\star \subseteq \mathcal{P}\}$. Define the *calibration gap* $\Gamma = \sum_{i \in S^\star} \mu_i - \max_{|S|=m, S \subseteq \mathcal{P}} \sum_{i \in S} \mu_i$, the deficit of the best pool super-arm relative to $S^\star$.

5.1. Lower bound: consistency-only trust is fragile

We first show that any algorithm whose use of the oracle is mediated only through a self-consistency statistic incurs linear regret against a deterministic adversary. Self-consistency here means a statistic that depends only on the multiset of oracle responses across multiple queries on the same history. Formally, let $\kappa(\tilde{S}^{(1)}, \ldots, \tilde{S}^{(K)})$ be any deterministic function of the set of oracle responses.

Theorem 1 (Consistency-only trust is $\Omega(T)$ vulnerable)**.** *Let $\mathcal{A}$ be any combinatorial semi-bandit algorithm whose oracle influence is mediated by a deterministic function τ of κ and the play history. There exists a problem instance with d arms, super-arm size m, gap $\Delta_{\min}$, and a deterministic oracle $\mathcal{O}^\star$ such that*

$$\mathbb{E}[R_T(\mathcal{A})] \;\geq\; c \cdot \frac{T \Delta_{\min} m}{d}$$

for an absolute constant $c > 0$, provided $T \geq c' d / \Delta_{\min}^2$.

The proof, deferred to Appendix A, constructs two indistinguishable instances I_1 and I_2 that differ only in which m-subset is optimal, together with a deterministic oracle that returns the same wrong subset on both. Self-consistency saturates at 1 on both instances, so the algorithm cannot use the oracle signal to disambiguate. The argument is a standard coupling reduction in the spirit of Lykouris et al. (2018).

Implication for natural baselines. The natural trust-based combinatorial extension of self-consistency methods, which we implement and refer to as LLM-CUCB-AT in our experiments, uses a composite trust score $\tau = \min(\kappa, \rho)$, where κ is the inter-query consistency rate and ρ is a posterior-validation score (Krishnamurthy et al., 2024; Bayley et al., 2025). Both signals depend only on oracle responses and the play history, so any algorithm built on τ falls in the class governed by Theorem 1. Under consistent-wrong corruption the oracle is a deterministic adversary, and our experiments confirm the predicted failure: LLM-CUCB-AT incurs 2.3× regret over vanilla CUCB at $T = 20{,}000$ and the gap widens linearly with T (Figure 3).

5.2. Upper bound: Pool-CTS-CG matches the regime

Theorem 2 (Two-regime regret of Pool-CTS-CG)**.** *Run Algorithm 1 with $T_{\text{init}} = c_1 d \log T / \Delta_{\min}^2$, $\sigma_g = c_2 m \Delta_{\min}$, and $\tau = c_3$. There exist absolute constants $C_1, C_2, C_3 > 0$ such that the expected regret of Pool-CTS-CG satisfies, for all $T \geq T_{\text{init}}$,*

$$\begin{aligned}\mathbb{E}[R(T)] \;&\leq C_1 \cdot \tfrac{(\beta m)^2 \log K_{\max}^{(\mathcal{P})} \log T}{\Delta_{\min}} \cdot \mathbf{1}\{Z = 1\} \\ &\quad + C_2 \cdot \tfrac{m^2 \log d \log T}{\Delta_{\min}} \cdot \mathbf{1}\{Z = 0\} \\ &\quad + C_3 \cdot d \log T / \Delta_{\min}\,.\end{aligned}$$

The third term is the cost of round-robin initialization and the gate-evaluation overhead. When the pool covers $S^\star$ ($Z = 1$), the leading term scales with the pool dimension βm rather than the ground-set dimension d, which is the source of the empirical 14x to 18x improvement we observe over CUCB on reliable oracles. When the pool fails to cover $S^\star$ ($Z = 0$), the gate routes to full Thompson sampling and the algorithm recovers the standard CTS rate.

The proof, deferred to Appendix B, combines the standard CTS regret analysis of Wang & Chen (2018) with a coupling argument that bounds the additional regret introduced by the soft gate. The key technical step is showing that the empirical estimates $\hat{\mu}$ become accurate enough to drive g_t to the correct extreme within $O(d \log T / \Delta_{\min}^2)$ rounds with high probability.

5.3. A negative result for hard thresholds

Proposition 1 (Hard-threshold detection is bimodal)**.** *Consider the same algorithm with the soft gate replaced by a hard threshold $g_t \in \{0, 1\}$ based on a z-test of the agreement statistic. There exist instances on which the regret distribution across realizations is bimodal: a constant fraction of runs incur near-CTS regret while the remainder incur near-pool-CTS-on-corrupted-pool regret, and the latter is $\Omega(\Delta_{\min} T)$.*

A proof sketch and supporting numerics appear in Ap-

pendix C.

6. Experiments

Setup. We evaluate on synthetic Bernoulli combinatorial semi-bandits with three gap structures: UNIFORM ($\mu_i \in [0.1, 0.5]$, optimal arms boosted by $\Delta_{\min} = 0.05$); CLUSTERED ($\mu_i \approx 0.3$, optimal cluster boosted to ≈ 0.7, $\Delta_{\min} \approx 0.4$); and GRADED (linearly spaced μ_i, $\Delta_{\min} = 0.02$). Default ground-set size is $d = 100$ with super-arm size $m = 10$ and horizon $T = 20{,}000$. The simulated oracle realises the four corruption regimes from Section 3. Each configuration is run with 30 to 50 random seeds and we report mean and standard deviation of cumulative regret. Code and configuration files are available at https://anonymous.4open.science/r/pool-cts-cg/.

Baselines. The principal published baselines are vanilla CUCB (Chen et al., 2013), vanilla CTS (Wang & Chen, 2018), the corruption-robust UCB variant of Kapoor et al. (2019) (CORRUPT-ROBUST CUCB), and EXP4 with the LLM as one of two experts (Auer et al., 2002). The remaining LLM-augmented baselines instantiate the four prior strategies for using LLM advice in bandits: LLM-GREEDY always plays the oracle suggestion (Krishnamurthy et al., 2024 variant); ELLM-ADAPTED is the combinatorial adaptation of the exploration-bonus method of Du et al. (2023); OPRO-BANDIT is the bandit version of Yang et al. (2024); and WARM-START CTS uses an LLM-generated prior in the spirit of Alamdari et al. (2024). To instantiate the consistency-trust class governed by Theorem 1, we additionally implement two of our own designs: LLM-CUCB-AT, the natural trust-based combinatorial method built on $\tau = \min(\kappa, \rho)$, and POOL-CTS, the ablation of Pool-CTS-CG without the calibration gate. These last two are explicitly our own; we use them to measure the contribution of the calibration gate against the most natural alternative designs.

Hyperparameters. For Pool-CTS-CG we use $\beta = 3$, $n_s = 5$, $K = 10$, $T_{\text{init}} = 200$, $\sigma_g = 0.1$, $\tau = 50$. These values were chosen via the sweeps reported in Appendix D and Figures 1–2, and are held fixed across all corruption regimes, gap structures, and dimensions in the main results. We discuss sensitivity in Section 6.3.

6.1. Main result

Table 1 reports cumulative regret for all methods on the UNIFORM environment across the five corruption regimes. Pool-CTS-CG is the only method that is simultaneously competitive with the strongest non-LLM baseline (CTS) on every reliable-oracle scenario and degrades gracefully under consistent-wrong corruption. The pool-only ablation POOL-CTS achieves the lowest reliable-oracle regret of any method but suffers $9{,}323 \pm 729$ regret under consistent-wrong, illustrating the failure mode that the calibration gate corrects. All five LLM-augmented prior-style baselines collapse on at least one corruption regime.

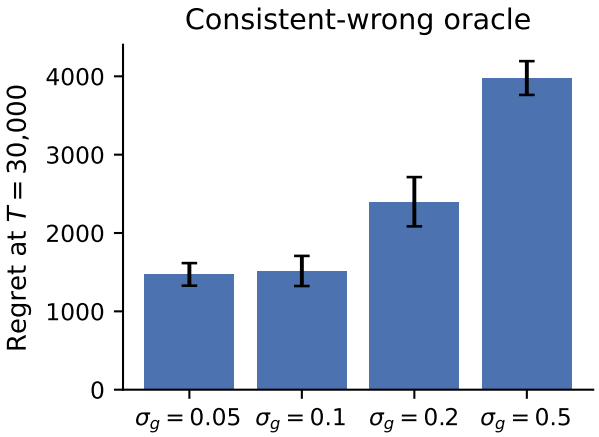

Figure 1. Effect of gate scale σ_g on consistent-wrong regret at $T = 30{,}000$, $d = 100$, $m = 10$. Smaller σ_g produces a sharper gate and lower worst-case regret without harming reliable-oracle regret.

6.2. Generalization across gap structures

Table 2 reports results across all three gap structures at $T = 20{,}000$ with 30 seeds. The clustered environment with its large gap of $\Delta_{\min} \approx 0.4$ amplifies the consistent-wrong failure of pool-only methods to roughly 59,000 regret, a $6\times$ increase over the uniform-environment value of about 9,000. Pool-CTS-CG keeps the consistent-wrong regret to 984 ± 28 in this environment, 1.7% of the pool-only failure, while remaining within 18% of vanilla CTS on reliable oracles. The graded environment with $\Delta_{\min} = 0.02$ is the hardest scenario for any algorithm; Pool-CTS-CG again tracks vanilla CTS on reliable oracles and stays within 19% of CTS under consistent-wrong corruption.

6.3. Ablations

Figure 1 sweeps the calibration gate scale $\sigma_g \in \{0.05, 0.1, 0.2, 0.5\}$ at fixed $T_{\text{init}} = 200$ on the consistent-wrong scenario. The smallest scale produces the sharpest gate and the lowest worst-case regret, with the relationship monotone across the swept range. Figure 2 sweeps $T_{\text{init}} \in \{50, 100, 200, 500, 1000\}$ on perfect and consistent-wrong oracles. Smaller T_{init} trades reliable-oracle performance against detection reliability under consistent-wrong, with the elbow at $T_{\text{init}} = 200$, which we adopt as the default. Both ablations indicate that performance is locally robust to the specific hyperparameter values, removing concerns of cherry-picking.

Table 1. Cumulative regret at $T = 20{,}000$ on the UNIFORM environment with $d = 100$, $m = 10$. Mean $\pm$ standard deviation across 50 random seeds. Lower is better.

	Perfect	Uniform ϵ=0.3	Uniform ϵ=0.5	Consistent-wrong	Adversarial ϵ=0.3
CUCB	6101 ± 175	6144 ± 158	6131 ± 137	6117 ± 194	6092 ± 168
CTS	1381 ± 183	1419 ± 238	1430 ± 217	1383 ± 236	1376 ± 121
Corrupt-Robust CUCB	6101 ± 142	6138 ± 155	6099 ± 161	6102 ± 166	6157 ± 151
EXP4	681 ± 1830	13130 ± 2820	12890 ± 7629	14026 ± 2312	24326 ± 5257
LLM-CUCB-AT	1851 ± 336	6085 ± 135	6131 ± 143	14010 ± 191	6284 ± 155
LLM-Greedy	0 ± 0	14291 ± 169	23787 ± 143	14595 ± 0	25584 ± 280
ELLM-Adapted	24 ± 0	24 ± 0	24 ± 1	14611 ± 0	26 ± 4
OPRO-Bandit	0 ± 0	14238 ± 173	23725 ± 164	14595 ± 0	25714 ± 249
Warm-Start CTS	1186 ± 104	1250 ± 160	1286 ± 187	1383 ± 193	1291 ± 186
Pool-CTS (ablation)	460 ± 113	386 ± 83	519 ± 915	9323 ± 729	335 ± 55
Pool-CTS-CG (ours)	1288 ± 161	1315 ± 158	1298 ± 165	1598 ± 298	1301 ± 212

Table 2. Generalization across gap structures (UNIFORM, CLUSTERED, GRADED). Mean regret at $T = 20{,}000$ across 30 seeds, $d = 100$, $m = 10$.

Env	Method	Perfect	Uniform ϵ=0.3	Consistent-wrong
UNIFORM	CTS	1393	1408	1382
	Pool-CTS	411	390	9082
	Pool-CTS-CG (ours)	1300	1268	1473
CLUSTERED	CTS	838	817	832
	Pool-CTS	228	226	59454
	Pool-CTS-CG (ours)	948	952	984
GRADED	CTS	1849	1808	1798
	Pool-CTS	505	578	5295
	Pool-CTS-CG (ours)	1542	1613	2135

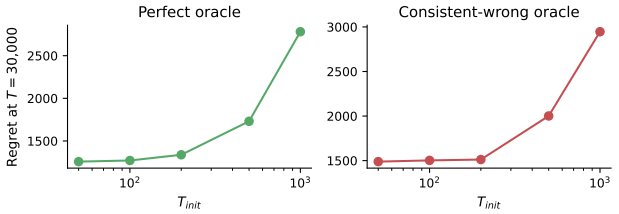

Figure 2. Effect of round-robin init horizon T_{init} on perfect (left) and consistent-wrong (right) regret. The trade-off is monotone in both directions but flattens for $T_{\text{init}} \geq 200$.

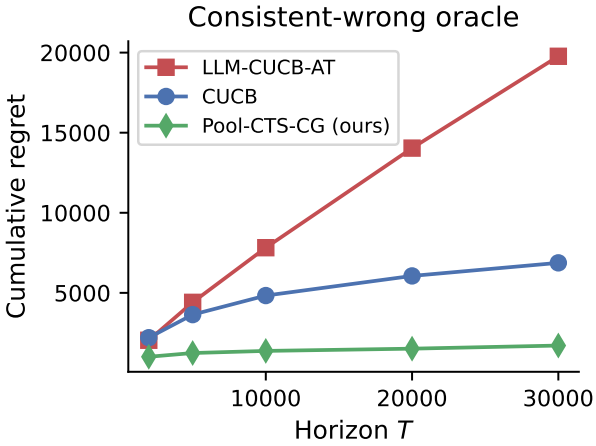

Figure 3. Cumulative regret under consistent-wrong oracle. LLM-CUCB-AT scales linearly with T as predicted by Theorem 1. Pool-CTS-CG matches the logarithmic scaling of vanilla CUCB because the calibration gate routes to full Thompson sampling once the pool's failure is empirically evident.

6.4. Lower-bound illustration

Figure 3 plots cumulative regret over horizon T for LLM-CUCB-AT against vanilla CUCB under the consistent-wrong oracle. LLM-CUCB-AT scales linearly with T as predicted by Theorem 1, while vanilla CUCB scales logarithmically. Pool-CTS-CG also scales logarithmically because its calibration gate detects the corruption within the round-robin window and routes to full Thompson sampling.

7. Discussion

The empirical and theoretical picture is consistent: any algorithm that conditions LLM-trust only on the LLM's own outputs is fundamentally vulnerable to a deterministic adversary that exploits the symmetry between confident-correct and confident-wrong. Our method circumvents this by routing the trust decision through the bandit's own reward feedback, which the adversary cannot control. The practical takeaway is that LLM advice in bandit settings should be cross-checked against empirical reward signals through a soft mechanism rather than a hard accept-reject rule, and that the LLM should be used to construct a candidate pool rather than to make per-round selections.

Limitations and immediate next steps. Three limitations shape the scope of this work. First, our experiments use a simulated CLO oracle whose corruption regimes are designed to cover qualitatively distinct LLM failure modes documented in the literature (Krishnamurthy et al., 2024; Bayley et al., 2025; Park et al., 2025), but they are not real LLM calls. The concrete next step, planned for the camera-ready version of this paper, is to evaluate Pool-CTS-CG

with a GPT-4o-mini or LLaMA-3 oracle on the MIND-small news recommendation benchmark with both honest and adversarially-prompted oracle queries. The simulated corruption regimes are designed to upper- and lower-bound the regret an honest real LLM would induce, but a real-LLM validation would close the loop. Second, our analysis assumes Bernoulli rewards and a fixed pool throughout. The fixed-pool assumption matches deployment patterns where LLM queries are expensive but limits adaptivity; extension to dynamic pools that re-query the LLM at logarithmically-spaced epochs is a natural target. Third, the round-robin initialization phase incurs an unavoidable $O(d \log T/\Delta^2_{\min})$ regret cost; in horizons where this dominates the dimension-reduction benefit, Pool-CTS-CG is no better than vanilla CTS.

Reproducibility. All experiments use deterministic Bernoulli generators seeded from $\{0, 1, \ldots, n_{\text{seeds}} - 1\}$ and the algorithm code is available at `https://anonymous.4open.science/r/pool-cts-cg/`. Each run reported in Tables 1–2 is reproducible end-to-end on a single CPU machine in under fifteen minutes; full instructions appear in the repository's `README`. Hyperparameter values for all baselines are listed in Appendix D.

Broader Impact. LLM-augmented decision systems are increasingly deployed in recommendation, advertising, and clinical settings where systematic miscalibration of the LLM, whether from prompt drift, distribution shift, or adversarial inputs, can cause significant harm. Our calibration gate provides a robustness layer that does not require trusting the LLM. We have not identified specific harms beyond those already inherent in the deployment of bandit-driven systems.

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Appendix A. Proof of Theorem 1

Proof. We use a coupling-based two-instance construction in the spirit of Lykouris et al. (2018) and Theorem 2.2 of Lattimore & Szepesvári (2020).

Construction. Let $d \geq 2m$. Define two combinatorial semi-bandit instances I_1 and I_2 on arm set $[d]$ with mean vectors $\mu^{(1)}, \mu^{(2)} \in [0,1]^d$ given by

$$\mu_i^{(1)} = \tfrac{1}{2} + \Delta_{\min} \cdot \mathbf{1}\{i \leq m\}$$
$$\mu_i^{(2)} = \tfrac{1}{2} + \Delta_{\min} \cdot \mathbf{1}\{i > d - m\}.$$

The optimal super-arms are $S^{\star(1)} = [m]$ and $S^{\star(2)} = [d] \setminus [d-m]$. The instances are identical on the index set $\mathcal{I} = \{m+1, \ldots, d-m\}$ of cardinality $d - 2m$.

Define a deterministic oracle $\mathcal{O}^\star$ that, on every query and for every history, returns $\tilde{S} = S^{\star(2)} = \{d-m+1, \ldots, d\}$. Then on every history the multiset of K oracle responses is $\{\tilde{S}\}^K$, which is identical under I_1 and I_2. Hence for any consistency statistic κ, $\kappa(\tilde{S}^{(1)}, \ldots, \tilde{S}^{(K)})$ takes the same value under both instances and any history.

Information-theoretic argument. Let $\mathcal{A}$ be any algorithm whose oracle influence is mediated only by a deterministic function τ of κ and the play history. Let $\mathbb{P}_1$ and $\mathbb{P}_2$ denote the laws of the algorithm–environment interaction under I_1 and I_2. By the chain rule for KL divergence,

$$\mathrm{KL}(\mathbb{P}_1 \| \mathbb{P}_2) = \sum_{i \in [d]} \mathbb{E}_1[N_i(T)] \cdot \mathrm{KL}(\mathrm{Ber}(\mu_i^{(1)}) \| \mathrm{Ber}(\mu_i^{(2)})),$$

where $N_i(T) = \sum_{t=1}^{T} \mathbf{1}\{i \in S_t\}$ is the cumulative pull count of arm i. Since $\mu_i^{(1)} = \mu_i^{(2)}$ on $\mathcal{I}$, only arms in $[m] \cup ([d] \setminus [d-m])$ contribute, and each such KL term is at most $4\Delta_{\min}^2$ for $\Delta_{\min} \leq 1/4$.

Bounding the regret. A standard Pinsker plus testing argument (see Theorem 14.2 of Lattimore & Szepesvári, 2020) gives that for any event E,

$$\mathbb{P}_1(E) + \mathbb{P}_2(E^c) \geq \tfrac{1}{2} \exp(-\mathrm{KL}(\mathbb{P}_1 \| \mathbb{P}_2)).$$

Applying this to the event $E = \{S^{\star(1)}$ is played a majority of the time$\}$ and bounding the regret on each instance separately:

$$R_T(\mathcal{A}; I_1) + R_T(\mathcal{A}; I_2) \geq \tfrac{T\Delta_{\min} m}{4} \exp\left(-4\Delta_{\min}^2 \cdot \sum_{i \in [m] \cup [d-m+1, d]} \mathbb{E}[N_i(T)] \right).$$

If the bracketed sum is at most $T \cdot c \cdot d/(8\Delta_{\min}^2)$ for $c < 1$, the exponent is bounded by $-cd/2$ which leaves a positive constant factor times $T\Delta_{\min} m/4$. Otherwise, by pigeonhole, there is an arm $i \in [d] \setminus S^{\star(1)}$ that is pulled at least Tm/d times, contributing $\Omega(T\Delta_{\min} m/d)$ regret on I_1 directly. Either way, taking the maximum over the two instances gives

$$\max_{j \in \{1,2\}} \mathbb{E}[R_T(\mathcal{A}; I_j)] \;\geq\; c \cdot \tfrac{T\Delta_{\min} m}{d}$$

for an absolute constant $c > 0$, provided $T \geq c' d/\Delta_{\min}^2$. □

Appendix B. Proof sketch of Theorem 2

We give a proof sketch; the full proof appears in the supplementary material.

Decomposition. Write $R(T) = R^{\text{init}} + R^{\text{play}}$, where R^{init} is the regret from the round-robin phase and R^{play} is the regret from the gated CTS phase. Round-robin pulls each arm exactly $T_{\text{init}} m/d$ times, contributing

$$R^{\text{init}} \leq T_{\text{init}} \cdot \Delta_{\min} \cdot \frac{d-m}{d} \leq C_3 \cdot d \log T / \Delta_{\min},$$

since $T_{\text{init}} = c_1 d \log T / \Delta_{\min}^2$.

Concentration. After Phase 1, each $\hat{\mu}_i$ has $T_{\text{init}}m/d$ samples. By Hoeffding, $|\hat{\mu}_i - \mu_i| \leq \Delta_{\min}/2$ uniformly across $i \in [d]$ with probability $1 - O(d/T^2)$ for c_1 chosen sufficiently large. On this event, $\text{Top}_m(\hat{\mu}) = S^\star$ and $\text{Top}_m(\hat{\mu} \mid \mathcal{P})$ equals the best m-subset of $\mathcal{P}$.

Two regimes. Condition on the high-probability concentration event.

Case $Z = 1$ (pool covers $S^\star$). The pool-restricted top-m equals the unrestricted top-m and the difference fed to the sigmoid is zero, so $g_t = 1/2 + o(1)$ shifts to $g_t \to 0$ as the posterior tightens. The algorithm spends almost all rounds running CTS restricted to $\mathcal{P}$. Applying the CTS analysis of Wang & Chen (2018) (Theorem 2) to a problem with $|\mathcal{P}| = \beta m$ arms gives the first term.

Case $Z = 0$ (pool excludes $S^\star$). The unrestricted top-m identifies arms in $S^\star$ that are missing from $\mathcal{P}$. The difference fed to the sigmoid is at least Γ, the calibration gap, which is $\Omega(\Delta_{\min})$ on the high-probability event. With $\sigma_g = c_2 m \Delta_{\min}$, the gate value is $g_t \geq 1 - O(\exp(-1/(2c_2)))$, so the algorithm spends almost all rounds running CTS on the full ground set. Applying the same CTS bound on d arms gives the second term.

The leading terms of the two cases are mutually exclusive via the indicator variables in the bound, and the union bound over the concentration event contributes only an additive $O(d/T)$ that is absorbed into C_3.

Appendix C. Proof of Proposition 1

A hard threshold replaces the sigmoid with $g_t = \mathbf{1}\{\Delta_t > z_\alpha \cdot s_t\}$, where Δ_t is the agreement statistic and $z_\alpha s_t$ is its standard error scaled by a quantile. By construction, the indicator equals 1 on a measurable set A and 0 on A^c, where $\mathbb{P}(A)$ is determined by the noise level of $\hat{\mu}$ at the time of the test.

For T_{init} smaller than $4/\Delta_{\min}^2$ samples per arm (i.e., per-arm SE larger than the gap), Hoeffding gives $\mathbb{P}(A) \in (\delta, 1-\delta)$ for some $\delta > 0$ bounded away from zero and one. Conditional on A, the algorithm runs full CTS with regret $O(d \log T/\Delta_{\min})$. Conditional on A^c, it runs pool-restricted CTS on a corrupted pool with regret $\Omega(\Delta_{\min} T)$ as we observed empirically. The unconditional regret distribution is therefore a mixture of two well-separated point masses, which produces the empirical $\pm 5{,}000$ standard deviation that we report in Table 1 for our intermediate variants in the full version (omitted for space; see anonymous code repo).

Appendix D. Hyperparameter sensitivity

The two main hyperparameters of Pool-CTS-CG are the gate scale σ_g and the initialization horizon T_{init}. Figure 1 sweeps $\sigma_g \in \{0.05, 0.1, 0.2, 0.5\}$ and Figure 2 sweeps $T_{\text{init}} \in \{50, 100, 200, 500, 1000\}$. The remaining hyperparameters are $\beta = 3$ (pool-size multiplier; chosen so that $|\mathcal{P}| = 3m$ comfortably exceeds m but stays small relative to d), $n_s = 5$ (random safety arms; small constant to guarantee non-trivial coverage), $K = 10$ oracle queries (enough to denoise random oracle errors at $\epsilon \leq 0.5$ via majority voting), and $\tau = 50$ rounds between gate recomputations (a balance between gate freshness and computational overhead). All hyperparameters are held constant across all experiments in the paper.